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Proteomic Profiling of *Deinococcus radiodurans* Reveals Irradiation-Induced Proteins and Their Associated Functional Pathways

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Abstract. *Deinococcus radiodurans* is an extremophilic microorganism renowned for its extraordinary resistance to ionizing radiation. Its unique DNA repair capacity and robust protein protection mechanisms make it an important model organism for studying cellular responses to radiation stress. Although multiple radiation-resistance mechanisms have been elucidated, proteomic-level insights—particularly regarding proteins specifically expressed after irradiation—remain limited. In this study, we employed high-resolution liquid chromatography-tandem mass spectrometry (LC-MS/MS) combined with bioinformatic analysis to systematically investigate the proteomic profiles of *D. radiodurans* at 0, 1, and 3 hours following exposure to 6 kGy of γ -radiation. Unlike traditional differential expression analysis, this study focused on proteins exclusively identified in the irradiated group, from which a total of 62 irradiation-induced proteins were identified. Gene Ontology (GO) enrichment analysis revealed that these proteins are predominantly involved in essential biological processes such as DNA repair, DNA binding, and cellular response to gamma radiation. Further analysis showed that irradiation-induced DNA repair proteins—RuvC, DdrA, and DdrB—exhibited a time-dependent increase in peptide-spectrum match (PSM) counts between 0 and 3 hours post-irradiation, indicating their critical roles in responding to radiation-induced genomic damage. These findings provide new proteomic evidence for understanding microbial radioresistance and offer promising molecular targets for applications in space biology and radiation protection technologies.

1. Introduction

Deinococcus radiodurans is one of the most radiation- and oxidation-resistant microorganisms known to date. It is capable of maintaining genomic and proteomic integrity under extreme conditions such as high-dose ionizing radiation and intense oxidative stress, thereby exhibiting remarkable environmental adaptability [1–4]. This extraordinary resistance makes it a powerful model organism for investigating radiation damage response mechanisms, developing radiation protection strategies, and exploring the potential for life in extraterrestrial environments. To date, numerous studies have focused on the DNA repair mechanisms [5–8] and antioxidant enzyme systems [1,9–11] of *D. radiodurans*. However, relatively few investigations have explored the proteins that are specifically induced after irradiation—i.e., proteins absent from unirradiated controls—and their functional roles in stress response.

To address this gap, the present study employs high-resolution liquid chromatography-tandem mass spectrometry (LC-MS/MS) coupled with bioinformatic analysis to systematically characterize the proteomic profiles of *D. radiodurans* at 0, 1, and 3 hours following exposure to 6



kGy γ -radiation. Distinct from conventional differential expression analyses, this work focuses specifically on proteins uniquely detected in irradiated samples, particularly those associated with key biological pathways such as DNA repair and radiation response (e.g., DdrA, DdrD, RuvC). By analyzing the qualitative presence and functional enrichment of these irradiation-induced proteins, this study aims to elucidate the early-stage molecular response mechanisms and potential regulatory pathways activated by radiation exposure. Overall, the findings provide not only new insights into the molecular basis of *D. radiodurans*' extreme resistance but also broader implications for microbial survival strategies, radiation protection, and space biology.

2. Materials and Methods

2.1. Strains and Culture Conditions

The strain *Deinococcus radiodurans* (catalog number CGMCC 1.633) was sourced from the China General Microbiological Culture Collection Center (CGMCC, Beijing, China). Bacterial cells were propagated in TGY broth comprising 1% tryptone, 0.5% glucose, and 0.1% yeast extract. Cultivation was carried out at 30 °C under constant shaking at 150 rpm.

2.2. Irradiation Conditions and Postirradiation Recovery

Bacterial cultures were grown at 30 °C and 150 rpm until they reached early stationary phase ($OD_{600} \approx 1.5$). Subsequently, they were exposed to ^{60}Co γ -radiation at a cumulative dose of 6 kGy (dose rate: 30 Gy/min) at Peking University (Beijing, China). Parallel cultures that were not irradiated served as controls.

After irradiation, both treated and control samples were centrifuged ($10,000 \times g$, 5 min, 4 °C), then resuspended in fresh TGY medium to a starting OD_{600} of 0.1. Cells were allowed to recover at 30 °C and 150 rpm, and samples were taken at 0, 1, 3, 6, and 12 hours post-irradiation. Each sample underwent two PBS washes, followed by centrifugation ($10,000 \times g$, 10 min, 4 °C). All procedures were performed in three independent biological replicates.

2.3. Protein Extraction and Digestion

Harvested cell pellets were lysed using a buffer containing 8 M urea, 2 mM EDTA, and 1 mM PMSF. Lysis was facilitated via probe sonication on ice (VCX800, 800 W, 20 kHz, pulse mode: 2 s on/2 s off for 4 min). The lysates were clarified by centrifugation ($15,000 \times g$, 45 min, 4 °C), and protein concentrations were determined using a BCA kit (Pierce, MA, USA).

For enzymatic digestion, 50 μg of protein from each sample was reduced with 10 mM DTT at 56 °C for 30 minutes, then alkylated in the dark using 50 mM IAA at room temperature for 30 minutes. To lower the urea concentration below 2 M, samples were diluted with 50 mM ammonium bicarbonate. Trypsin digestion was performed overnight (enzyme-to-substrate ratio of 1:50, w/w) at 37 °C, followed by an additional 4-hour digestion at a 1:100 ratio.

The reaction was terminated using 1% formic acid (v/v). Resulting peptides were desalted via Monospin C18 cartridges (Shimadzu, Japan), vacuum-dried, and stored at -80 °C for further analysis.

2.4. NanoLC-MS/MS Analysis

Dried peptides were reconstituted in 0.1% formic acid and centrifuged ($12,000 \times g$, 10 min) before LC-MS/MS analysis. Peptides were analyzed using an Easy-nLC 1000 system coupled to a Q-Exactive HFX mass spectrometer (Thermo Fisher Scientific, Bremen, Germany). Separation was achieved using a homemade C18 reverse-phase column (100 μm ID $\times$ 15 cm, 1.9 μm resin), using a 90-minute gradient at 300 nL/min flow rate.

The mobile phases were: A (0.1% formic acid in water) and B (0.1% formic acid in 80% acetonitrile). The gradient schedule was: 5% B at 0 min, 10% at 16 min, 22% at 60 min, 35% at 78 min, increasing to 99% at 83–85 min, and re-equilibrating to 5% from 86–90 min.

Mass spectrometry data were acquired in DDA mode, scanning the m/z range of 350–1500 with a 0.25 s accumulation time. The top 20 most intense ions were selected for MS/MS, with dynamic exclusion enabled for 20 seconds.

2.5. Protein Identification

The mass spectrometry raw files were processed using pFind 3 (v3.2.2) to identify proteins [12]. Searches were carried out against the *Deinococcus radiodurans* proteome (UniProtKB: UP000002524, downloaded on October 2, 2019). Parameters included: precursor and fragment mass tolerances of 20 ppm, trypsin digestion with up to three missed cleavages, and Open Search mode enabled. Carbamidomethylation (C) was treated as a fixed modification, and methionine oxidation (M) as a variable one.

2.6. Bioinformatic Analysis

Proteins identified using pFind3 were subjected to Gene Ontology (GO) functional enrichment analysis using DAVID Bioinformatics Resources 6.8, including the categories of biological process (GO_BP), cellular component (GO_CC), and molecular function (GO_MF). The enrichment threshold was set with an EASE score cutoff of 0.05 and a minimum count of 2 for each GO term. Data processing and statistical analyses were performed using Python, and figures were created with BioRender (<https://BioRender.com>).

3 Results and discussion

To systematically investigate the proteomic changes in *Deinococcus radiodurans* at different recovery stages following γ -radiation exposure, we established the proteomic workflow illustrated in Figure 1. Bacterial cultures were exposed to 6 kGy of γ -radiation and subsequently allowed to recover. Samples were collected at 0, 1, and 3 hours post-irradiation. Proteins were extracted, enzymatically digested, and analyzed via LC-MS/MS. Protein identification was performed using the pFind software, followed by bioinformatic analysis to annotate functional categories and assess biological significance.

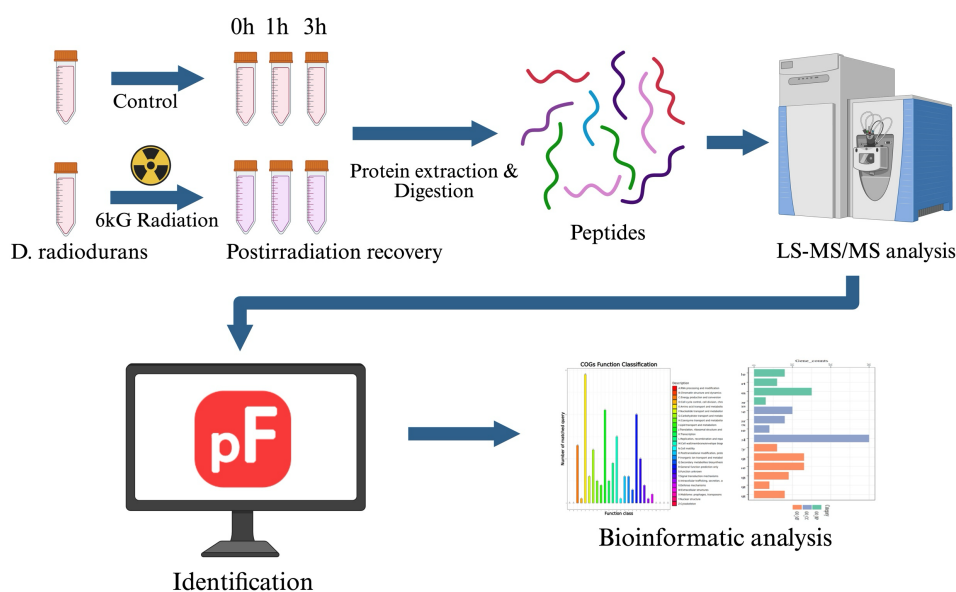


Figure 1. Experimental workflow for proteomic profiling of *Deinococcus radiodurans* under radiation and recovery conditions. Cultures of *Deinococcus radiodurans* were divided into a control group and a 6 kGy γ -irradiation group. Samples were collected at 0 h, 1 h, and 3 h post-irradiation recovery (PIR). Proteins were extracted and digested into peptides for LC-MS/MS analysis. The resulting spectra were processed using the pFind software for protein identification, followed by functional enrichment and classification using bioinformatic tools.

As shown in Figure 2a, a substantial number of peptide-spectrum matches (PSMs) were detected in both irradiated and control groups across all three time points, with the number of PSMs gradually decreasing over time. Notably, the irradiated group consistently exhibited PSM counts comparable to or slightly higher than the control group, suggesting that *D. radiodurans* maintains a robust capacity to rapidly initiate protein synthesis following intense radiation stress. This observation aligns with previous reports describing its highly efficient protein protection systems and strong resistance to redox damage [1,13,14].

In terms of overall protein identification (Figure 2), approximately 2,000 proteins were consistently detected at each time point. Comparative analysis of the proteomic composition between groups (Figure 2b) revealed that 2,034 proteins were shared between irradiated and control samples, while 142 proteins were uniquely detected in the control group and 62 proteins were exclusive to the irradiated group. These findings indicate that although a core set of proteins

is stably expressed under both conditions, radiation exposure triggers the induction of a distinct subset of proteins with potential functional relevance.

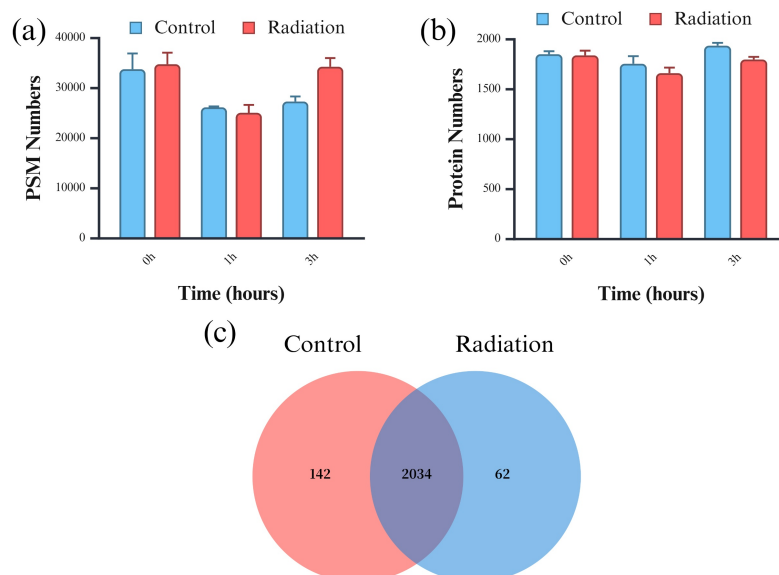


Figure 2. Overview of peptide-spectrum matches and protein identification across control and radiation groups **(a)** The number of peptide-spectrum matches (PSMs) identified by pFind in control (blue) and radiation (red) groups at 0 h, 1 h, and 3 h post-irradiation recovery (PIR). **(b)** The number of proteins identified by pFind in control (blue) and radiation (red) groups at 0 h, 1 h, and 3 h PIR. **(c)** Venn diagram showing the overlap and unique proteins identified in the control and radiation groups.

To further explore the functional significance of these group-specific proteins, we performed Gene Ontology (GO) enrichment analysis on proteins exclusively identified in each group (Figure 3). In the control group (Figure 3a), the unique proteins were primarily enriched in categories such as "transmembrane transporter activity," "pseudouridine synthase activity," and "regulation of intracellular pH," suggesting a role in homeostasis and molecular transport. In contrast, proteins uniquely expressed in the irradiated group (Figure 3b) were significantly enriched in pathways including "DNA binding," "DNA repair," and "cellular response to gamma radiation," indicating that DNA damage response pathways were specifically activated upon radiation exposure.

To gain further insights into the temporal dynamics of irradiation-induced DNA repair proteins, we analyzed the PSM counts of three key proteins—RuvC, DdrA, and DdrB—across recovery time points (**Figure 4**). All three proteins were exclusively identified in the irradiated

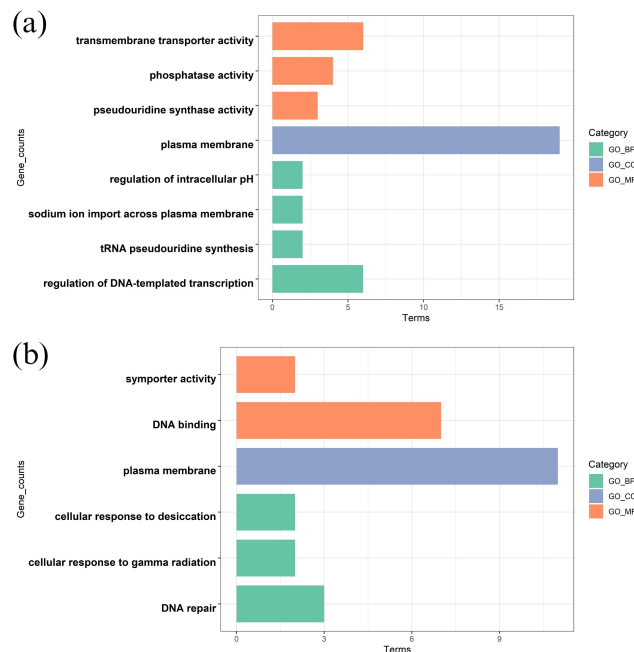


Figure 3. GO enrichment analysis of group-specific proteins identified in control and radiation groups. **(a)** GO terms enriched by proteins uniquely identified in the control group. **(b)** GO terms enriched by proteins uniquely identified in the radiation group.

group, and their detection frequency increased progressively over time. In particular, DdrA exhibited a marked increase in expression from 0 to 3 hours post-irradiation.

RuvA, together with RuvB and RuvC, forms the RuvA–RuvB–RuvC complex that processes Holliday junction (HJ) DNA during genetic recombination and DNA repair [15]. DdrA is known to bind DNA ends and protect them from exonuclease degradation, making it a critical factor for maintaining genome integrity under extreme radiation conditions [16–18]. DdrB functions in RecA-independent DNA repair and genome reconstruction, playing a pivotal role in reassembling severely fragmented DNA following high-dose irradiation [16,19,20]. The exclusive identification

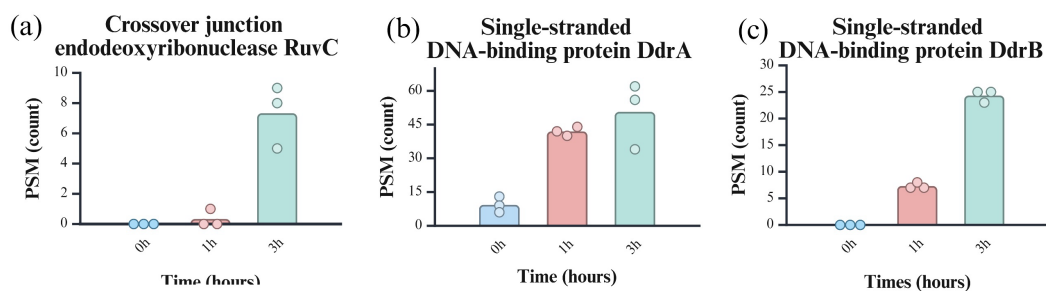


Figure 4. PSM counts of DNA damage response- and repair-related proteins after radiation. **(a)**-**(c)** PSM counts of RuvC, DdrA, and DdrB—proteins identified exclusively in the radiation group.

and radiation-dependent accumulation of these proteins in the irradiated group underscores their central roles in *D. radiodurans*' radiation-induced stress response.

4 Conclusion

In this study, high-resolution mass spectrometry combined with bioinformatic analysis was employed to systematically characterize the proteomic changes in *Deinococcus radiodurans* within 0–3 hours after exposure to 6 kGy of γ -radiation. Particular emphasis was placed on proteins that were exclusively expressed in the irradiated group, revealing their involvement in DNA repair and radiation response pathways. Although a large number of core proteins were shared between the control and irradiated samples, radiation treatment specifically induced the expression of 62 unique proteins. GO enrichment analysis showed that these proteins were significantly associated with functional categories such as "DNA repair," "DNA binding," and "cellular response to gamma radiation." Further analysis revealed that key DNA repair proteins—RuvC, DdrA, and DdrB—accumulated progressively during the recovery period and were exclusively detected in the irradiated group, indicating their essential roles in the rapid response to radiation-induced damage and the maintenance of genome integrity. Overall, these findings enhance our understanding of the molecular basis underlying the extraordinary radioresistance of *D. radiodurans*, and provide novel insights into microbial survival strategies, radiation protection, and space biology.

References

- [1] Slade D, Radman M. Oxidative Stress Resistance in *Deinococcus radiodurans*. *Microbiol Mol Biol Rev* 2011;75:133–91. <https://doi.org/10.1128/MMBR.00015-10>.
- [2] Krisko A, Radman M. Biology of Extreme Radiation Resistance: The Way of *Deinococcus radiodurans*. *Cold Spring Harbor Perspectives in Biology* 2013;5:a012765–a012765. <https://doi.org/10.1101/cshperspect.a012765>.
- [3] Minton KW. DNA repair in the extremely radioresistant bacterium *Deinococcus radiodurans*. *Molecular Microbiology* 1994;13:9–15. <https://doi.org/10.1111/j.1365-2958.1994.tb00397.x>.
- [4] Sweet DM, Moseley BEB. Accurate repair of ultraviolet-induced damage in *Micrococcus radiodurans*. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis* 1974;23:311–8. [https://doi.org/10.1016/0027-5107\(74\)90104-3](https://doi.org/10.1016/0027-5107(74)90104-3).
- [5] Sharma DK, Misra HS, Soni I, Rajpurohit YS. Characterization of DNA Processing Protein A (DprA) of the Radiation-Resistant Bacterium *Deinococcus radiodurans*. *Microbiol Spectr* 2022;10:e0347022. <https://doi.org/10.1128/spectrum.03470-22>.
- [6] Cox MM, Battista JR. *Deinococcus radiodurans* — the consummate survivor. *Nat Rev Microbiol* 2005;3:882–92. <https://doi.org/10.1038/nrmicro1264>.
- [7] Slade D, Lindner AB, Paul G, Radman M. Recombination and Replication in DNA Repair of Heavily Irradiated *Deinococcus radiodurans*. *Cell* 2009;136:1044–55. <https://doi.org/10.1016/j.cell.2009.01.018>.
- [8] Repar J, Cvjetan S, Slade D, Radman M, Zahradka D, Zahradka K. RecA protein assures fidelity of DNA repair and genome stability in *Deinococcus radiodurans*. *DNA Repair* 2010;9:1151–61. <https://doi.org/10.1016/j.dnarep.2010.08.003>.
- [9] Daly MJ, Gaidamakova EK, Matrosova VY, Kiang JG, Fukumoto R, Lee D-Y, et al. Small-Molecule Antioxidant Proteome-Shields in *Deinococcus radiodurans*. *PLoS ONE* 2010;5:e12570. <https://doi.org/10.1371/journal.pone.0012570>.
- [10] Tian B, Xu Z, Sun Z, Lin J, Hua Y. Evaluation of the antioxidant effects of carotenoids from *Deinococcus radiodurans* through targeted mutagenesis, chemiluminescence, and DNA damage analyses. *Biochimica et Biophysica Acta (BBA) - General Subjects* 2007;1770:902–11. <https://doi.org/10.1016/j.bbagen.2007.01.016>.
- [11] Kobayashi I, Tamura T, Sghaier H, Narumi I, Yamaguchi S, Umeda K, et al. Characterization of monofunctional catalase KatA from radioresistant bacterium *Deinococcus radiodurans*. *Journal of Bioscience and Bioengineering* 2006;101:315–21. <https://doi.org/10.1263/jbb.101.315>.
- [12] Chi H, Liu C, Yang H, Zeng W-F, Wu L, Zhou W-J, et al. Comprehensive identification of peptides in tandem mass spectra using an efficient open search engine. *Nat Biotechnol* 2018;36:1059–61. <https://doi.org/10.1038/nbt.4236>.
- [13] Sadowska-Bartosz I, Bartosz G. Antioxidant defense of *Deinococcus radiodurans*: how does it contribute to extreme radiation resistance? *International Journal of Radiation Biology* 2023;99:1803–29. <https://doi.org/10.1080/09553002.2023.2241895>.
- [14] Han JM, Mwiti G, Yeom S, Lim J, Kim WS, Lim S, et al. Radiation-Resistant Bacteria *Deinococcus radiodurans* - Derived Extracellular Vesicles as Potential Radioprotectors. *Adv Healthcare Materials* 2024;2403192. <https://doi.org/10.1002/adhm.202403192>.
- [15] Sun Y, Yang J, Xu G, Cheng K. Biochemical and Structural Study of RuvC and YggF from *Deinococcus radiodurans*. *mBio* 2022;13:e0183422. <https://doi.org/10.1128/mbio.01834-22>.
- [16] Tanaka M, Earl AM, Howell HA, Park M-J, Eisen JA, Peterson SN, et al. Analysis of *Deinococcus radiodurans*'s Transcriptional Response to Ionizing Radiation and Desiccation Reveals Novel Proteins That Contribute to Extreme Radioresistance. *Genetics* 2004;168:21–33. <https://doi.org/10.1534/genetics.104.029249>.

- [17] Gao L, Zhou Z, Chen X, Zhang W, Lin M, Chen M. Comparative Proteomics Analysis Reveals New Features of the Oxidative Stress Response in the Polyextremophilic Bacterium *Deinococcus radiodurans*. *Microorganisms* 2020;8:451. <https://doi.org/10.3390/microorganisms8030451>.
- [18] Harris DR, Tanaka M, Saveliev SV, Jolivet E, Earl AM, Cox MM, et al. Preserving Genome Integrity: The DdrA Protein of *Deinococcus radiodurans* R1. *PLoS Biol* 2004;2:e304. <https://doi.org/10.1371/journal.pbio.0020304>.
- [19] Norais CA, Chitteni-Pattu S, Wood EA, Inman RB, Cox MM. DdrB protein, an alternative *Deinococcus radiodurans* SSB induced by ionizing radiation. *J Biol Chem* 2009;284:21402–11. <https://doi.org/10.1074/jbc.M109.010454>.
- [20] Xu G, Lu H, Wang L, Chen H, Xu Z, Hu Y, et al. DdrB stimulates single-stranded DNA annealing and facilitates RecA-independent DNA repair in *Deinococcus radiodurans*. *DNA Repair (Amst)* 2010;9:805–12. <https://doi.org/10.1016/j.dnarep.2010.04.006>.